# Answers: Introduction to probability

Michelle Arnetta

**Summary**

Answers to questions relating to the guide on introduction to probability.

*These are the answers to Questions: Introduction to probability.*

**Please attempt the questions before reading these answers!**

## Q1

(Note: accept all simplified fractions, or probabilities represented in decimal and percentage forms)

### 1.1.

(a)

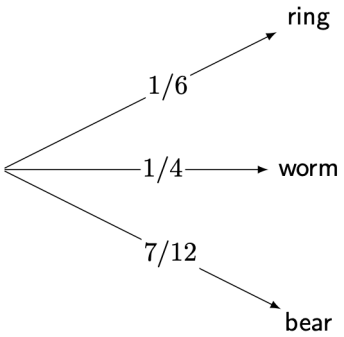



Figure 1: Answer to Q1.1(a)

(b)

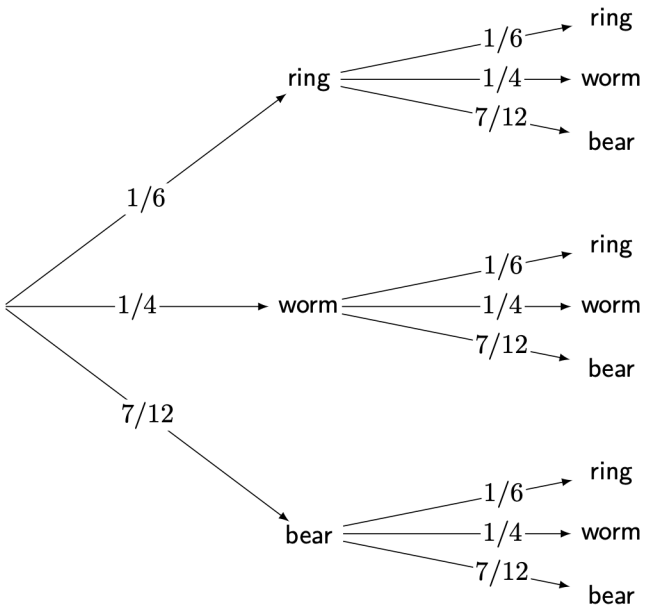



Figure 2: Answer to Q1.1(b)

(c)

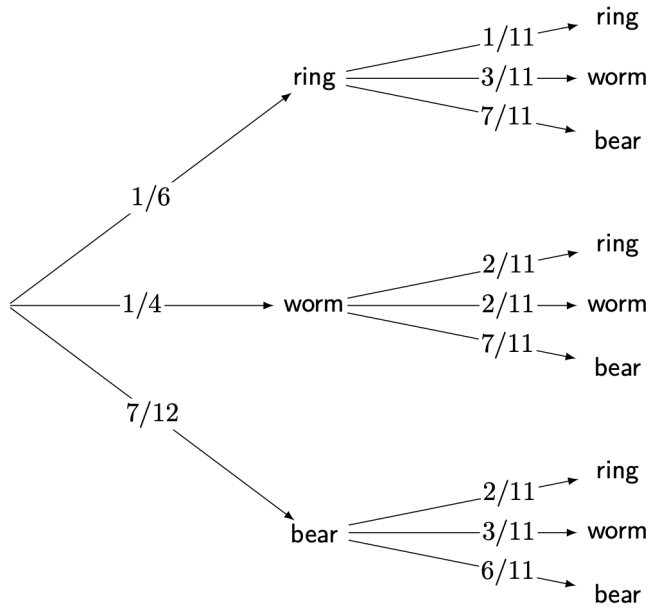



Figure 3: Answer to Q1.1(c)

**1.2.**

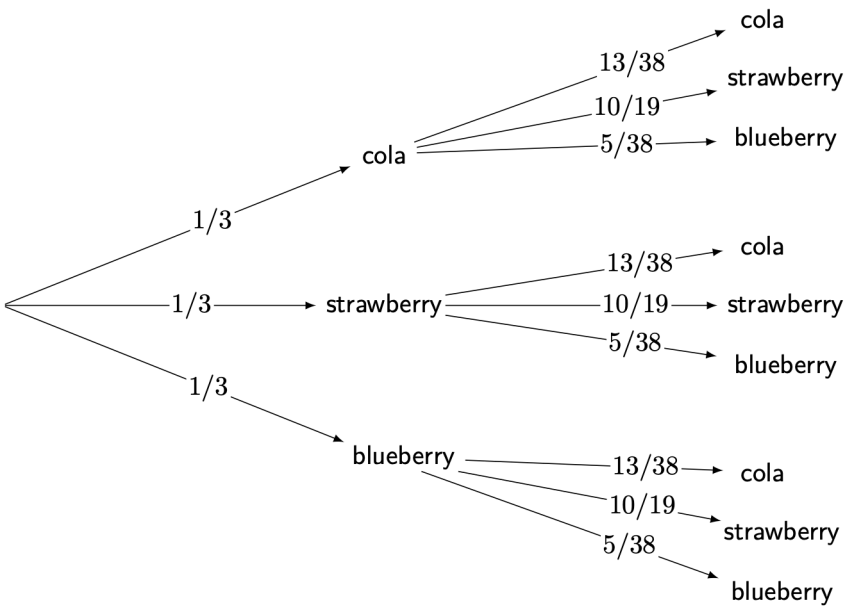



Figure 4: Answer to Q1.2

**1.3.**

- The events in Q1.1.(b) are independent.
- The events in Q1.1.(c) are dependent.
- The events in Q1.2. are independent.

---

# Q2

(Note: accept all simplified fractions, or probabilities represented in decimal and percentage forms)

**2.1.**

(a) $\mathbb{P}(\text{gummy bear}) = \frac{7}{12}$. Therefore, apply the complement rule to calculate the complement of $\mathbb{P}(\text{gummy bear})$, so $\mathbb{P}(\text{gummy bear}') = 1 - \frac{7}{12} = \frac{5}{12}$.

(b) The probability of drawing a gummy ring the first time is $\frac{2}{12}$, and the probability of drawing a gummy ring the second time is also $\frac{2}{12}$. Therefore, $\mathbb{P}(\text{gummy ring and gummy ring}) = (\frac{2}{12})(\frac{2}{12}) = \frac{4}{144} = \frac{1}{36}$.

(c) The probability of drawing a gummy bear the first time is $\frac{7}{12}$, and the probability of drawing a gummy worm the second time is $\frac{3}{11}$. Therefore, $\mathbb{P}(\text{gummy bear then gummy worm}) = (\frac{7}{12})(\frac{3}{11}) = \frac{21}{132} = \frac{7}{44}$.

**2.2.** The probability of drawing a cola flavored jelly bean the first time is $\frac{10}{30}$, and the probability of drawing a strawberry flavored jelly bean the second time is $\frac{20}{38}$. Therefore, $\mathbb{P}(\text{soda and strawberry}) = (\frac{10}{30})(\frac{20}{38}) = \frac{200}{1140} = \frac{10}{57}$.

---

## Q3

**3.1.** This is an example of experimental probability.

**3.2.** The total number of spins is 60, and it lands on white 17 times. Therefore, $\mathbb{P}(\text{white}) = \frac{17}{60}$, so by the complement rule, $\mathbb{P}(\text{white}') = 1 - \frac{17}{60} = \frac{43}{60}$.

**3.3.** The spinner is unbiased (meaning the probability is uniform), so there are four possible colors that the spinner is equally likely to land on. Calculating the theoretical probability, $\mathbb{P}(\text{red}) = \frac{1}{4}$.

**3.4.** As you spin the spinner more times, the experimental probabilities of each color will get closer to their theoretical probabilities.

---

## Q4

**4.1.** The sample space is $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$, which contains 20 possible outcomes. The subset of the sample space with a value above 12 is $\{13, 14, 15, 16, 17, 18, 19, 20\}$, which contains 8 out of the 20 possible outcomes. Therefore, $\mathbb{P}(\text{success}) = \frac{8}{20} = \frac{2}{5}$.

**4.2.** The possible outcomes of the 5-sided dice roll are $\{1, 2, 3, 4, 5\}$. Adding 3 to each number, the sample space becomes $\{(1{+}3), (2{+}3), (3{+}3), (4{+}3), (5{+}3)\} = \{4, 5, 6, 7, 8\}$. The sample space contains 5 possible outcomes, and the subset of the sample space that contains values 5 and above is $\{5, 6, 7, 8\}$, which contains 4 out of 5 possible outcomes. Therefore, $\mathbb{P}(x \geq 5) = \frac{4}{5}$ where $x$ points of damage are dealt to the dragon.

**4.3.** Here is the sample space represented as a table:

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **1** | (1,1) | (1,2) | (1,3) | (1,4) |
| **2** | (2,1) | (2,2) | (2,3) | (2,4) |

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **3** | (3,1) | (3,2) | (3,3) | (3,4) |
| **4** | (4,1) | (4,2) | (4,3) | (4,4) |

For convenience purposes, all outcomes that do not contain a 4 are then marked with an asterisk (although any way of marking or counting these outcomes are acceptable):

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **1** | (1,1)* | (1,2)* | (1,3)* | (1,4) |
| **2** | (2,1)* | (2,2)* | (2,3)* | (2,4) |
| **3** | (3,1)* | (3,2)* | (3,3)* | (3,4) |
| **4** | (4,1) | (4,2) | (4,3) | (4,4) |

There are 16 total possible outcomes in the sample space, and there are 9 outcomes that do not contain a 4. Therefore, $\mathbb{P}(\text{failure}) = \frac{9}{16}$.

**4.4.** Here is the sample space represented as a table:

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **1** | (1,1) | (1,2) | (1,3) | (1,4) |
| **2** | (2,1) | (2,2) | (2,3) | (2,4) |
| **3** | (3,1) | (3,2) | (3,3) | (3,4) |
| **4** | (4,1) | (4,2) | (4,3) | (4,4) |
| **5** | (5,1) | (5,2) | (5,3) | (5,4) |
| **6** | (6,1) | (6,2) | (6,3) | (6,4) |
| **7** | (7,1) | (7,2) | (7,3) | (7,4) |
| **8** | (8,1) | (8,2) | (8,3) | (8,4) |
| **9** | (9,1) | (9,2) | (9,3) | (9,4) |

The results of the two dice rolls can then be added together:

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **1** | 2 | 3 | 4 | 5 |

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **2** | 3 | 4 | 5 | 6 |
| **3** | 4 | 5 | 6 | 7 |
| **4** | 5 | 6 | 7 | 8 |
| **5** | 6 | 7 | 8 | 9 |
| **6** | 7 | 8 | 9 | 10 |
| **7** | 8 | 9 | 10 | 11 |
| **8** | 9 | 10 | 11 | 12 |
| **9** | 10 | 11 | 12 | 13 |

For convenience purposes, all outcomes with skill levels that are greater than 9 are then marked with an asterisk (although any way of marking or counting these outcomes are acceptable):

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **1** | 2 | 3 | 4 | 5 |
| **2** | 3 | 4 | 5 | 6 |
| **3** | 4 | 5 | 6 | 7 |
| **4** | 5 | 6 | 7 | 8 |
| **5** | 6 | 7 | 8 | 9 |
| **6** | 7 | 8 | 9 | 10* |
| **7** | 8 | 9 | 10* | 11* |
| **8** | 9 | 10* | 11* | 12* |
| **9** | 10* | 11* | 12* | 13* |

There are 36 total possible outcomes in the sample space, and there are 10 outcomes where the skill level exceeds 9 points. Therefore, $\mathbb{P}(x > 9) = \frac{10}{36} = \frac{5}{18}$ where $x$ is the number of skill level points.

## Version history and licensing

v1.0: initial version created 04/25 by Michelle Arnetta as part of a University of St Andrews VIP project.

This work is licensed under CC BY-NC-SA 4.0.